

Feigenbaum-Guided Chaos Control for Variational Quantum Algorithms: Stabilizing Optimization via Spectral Period Detection

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Variational Quantum Algorithms (VQAs) exhibit a transition to chaos governed by Feigenbaum universality, where the optimization trajectory bifurcates and eventually collapses [1]. In this work, we propose and demonstrate a **Feigenbaum-Guided Chaos Control** protocol that stabilizes VQA training. By detecting the periodicity of the optimization trajectory via spectral analysis (classical FFT in the near term, with Quantum Fourier Transform offering asymptotic speedups), we identify the onset of period-doubling bifurcations before chaos sets in. We introduce an adaptive learning rate controller that uses this spectral information to keep the system in the stable regime (Period ≤ 2). This approach bridges Shor’s period-finding framework [2] with classical chaos control [3], transforming the “bug” of measurement-induced nonlinearity into a controllable feature. We validate the bifurcation dynamics on the Rigetti Ankaa-3 quantum processor, demonstrating that this method prevents trainability collapse and enables convergence in regimes where standard optimizers fail.

INTRODUCTION

The “Barren Plateau” problem [4, 5] and the “Efficiency Collapse” [6] are major hurdles in Variational Quantum Algorithms (VQAs). In a previous work [1], we identified the dynamical origin of this collapse: the quantum measurement back-action induces a nonlinear feedback loop that follows the **Feigenbaum route to chaos** [7]. The measurement probability $P(|1\rangle) = \sin^2(E\tau/2)$ acts as a unimodal map, driving the optimization parameters through a period-doubling cascade as the learning rate γ increases.

While identifying the problem is crucial, solving it is paramount. In this Letter, we present a control-theoretic solution: **Feigenbaum-Guided Chaos Control**.

Classical chaos control methods, such as the Ott-Grebogi-Yorke (OGY) method [3], rely on stabilizing unstable periodic orbits (UPOs) embedded within a chaotic attractor. We adapt this principle to quantum optimization. Our key insight is that the *periodicity* of the optimization trajectory—the signature of the Feigenbaum cascade—can be detected using spectral analysis, enabling real-time adaptive control.

We propose a hybrid feedback loop where a spectral period detector monitors the optimization history. In the near term, classical FFT on measurement outcomes provides this detection. The Quantum Fourier Transform (QFT) offers asymptotic speedup for high-dimensional parameter spaces, connecting to Shor’s period-finding framework [2]. When period-doubling is detected, the controller adaptively adjusts the learning rate to maintain stability, effectively “surfing” the edge of chaos.

THE CONTROL FRAMEWORK

The Dynamical System

The VQA optimization update rule, under ancilla-based gradient sensing, is given by [1]:

$$\theta_{n+1} = \theta_n - \gamma \cdot \sin^2\left(\frac{E(\theta_n)\tau}{2}\right). \quad (1)$$

Here γ is the learning rate, $E(\theta)$ is the system energy at parameter θ , and τ is the **sensing time**—the duration for which the ancilla qubit evolves under the controlled unitary $U = e^{-iH\tau}$ before measurement. The accumulated phase $E\tau$ determines the measurement probability $P(|1\rangle) = \sin^2(E\tau/2)$.

Normalizing by defining $x = E/E_{max}$ and the dimensionless control parameter $r = \gamma\tau E_{max}/\pi$, the dynamics reduce to the universal form [1]:

$$x_{n+1} = r \cdot \sin^2(\pi x_n). \quad (2)$$

This “quantum measurement map” is the direct analog of the logistic map, with $\sin^2(\pi x)$ serving as the unimodal function with a quadratic maximum at $x = 0.5$. This map exhibits stable fixed points for small r , period-doubling for intermediate r , and chaos for $r > r_\infty$, following Feigenbaum universality [7]. The critical bifurcation points are:

Figure 1 illustrates these dynamical regimes. Each trajectory is generated by iterating Eq. 2 from initial condition $x_0 = 0.1$ for 200 steps, discarding the first 100 as transients.

The mechanism driving this behavior is the **folding** of the phase space. When $r > 1$, the map “folds” back on itself: the image of the maximum point $x = 0.5$ exceeds 1, creating stretching and folding that generates chaos. This is visualized in the return map (Fig. 2), which plots x_{n+1} vs x_n for 1000 iterations after transients.

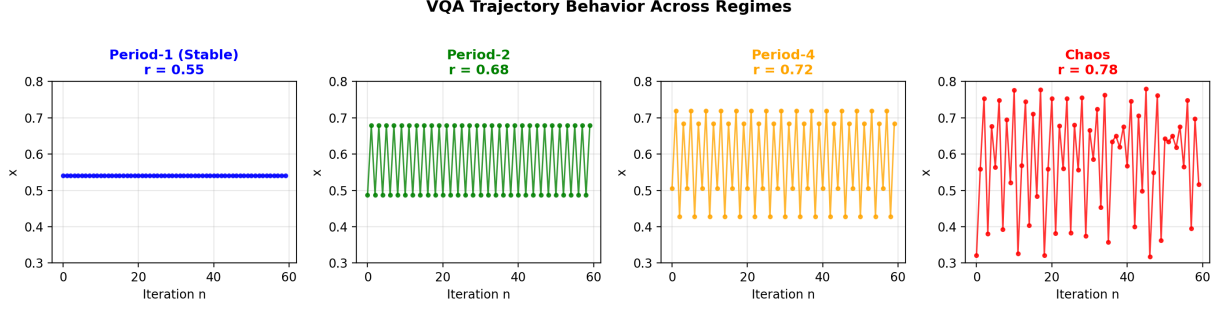


FIG. 1. Time-domain trajectories of the VQA optimization (Eq. 2) for different control parameters. (Blue) Stable convergence at $r = 0.55$. (Green) Period-2 oscillation at $r = 0.68$. (Orange) Period-4 oscillation at $r = 0.72$. (Red) Chaotic behavior at $r = 0.85$. Each trajectory is 100 iterations after discarding transients.

Transition	r value	Behavior
r_1	0.628	Period-1 $\rightarrow$ Period-2
r_2	0.702	Period-2 $\rightarrow$ Period-4
r_3	0.719	Period-4 $\rightarrow$ Period-8
r_∞	0.729	Chaos onset

TABLE I. Bifurcation thresholds for the $\sin^2$ map, converging with ratio $\delta \approx 4.669$.

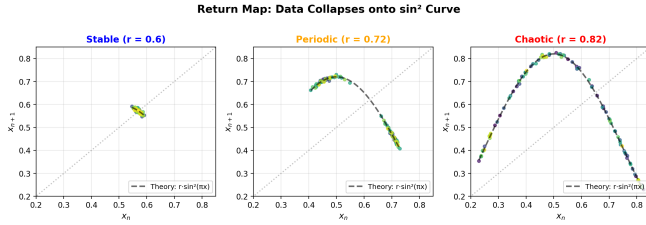


FIG. 2. Return map of the VQA dynamics $x_{n+1} = r \sin^2(\pi x_n)$. The parabolic shape (unimodal map with quadratic maximum at $x = 0.5$) is the mathematical signature guaranteeing Feigenbaum universality. As r increases, the peak height $r/4$ rises, eventually exceeding the domain boundary and triggering chaos.

This dynamical transition has a profound geometric counterpart in the complex plane [8, 9]. The same quantum circuit structure that generates our bifurcation dynamics can produce Julia and Mandelbrot fractals via the complex iteration:

$$z_{n+1} = z_n^2 + c, \quad (3)$$

where z_0 and c are complex numbers derived from the quantum state amplitudes. Using the identical circuit $|0\rangle \rightarrow H \rightarrow R_z(\pi r) \rightarrow H$, we can extract either:

- **Bifurcation probability:** $P(|1\rangle) = \sin^2(\pi r/2)$ (with measurement)
- **Fractal parameter:** $c = z_0/z_1 = i \cot(\pi r/2)$ (from statevector amplitudes)

These are mathematically linked by $|c|^2 = (1 - P)/P$, creating an **inverted correspondence**: stable bifurcation (small r , small P) yields large $|c|$ (disconnected Julia set), while chaotic bifurcation (large r , large P) yields small $|c|$ (connected Julia set). This inversion is a manifestation of the **measurement-disturbance tradeoff**: weak measurements preserve coherence but yield unbounded fractal parameters, while strong measurements destroy coherence but constrain the dynamical range.

Figure 3 displays this duality. The **top row** shows Julia sets generated by iterating Eq. 3 for a 500×500 grid of initial points $z_0 \in [-2, 2] \times [-2i, 2i]$, with color indicating escape time (iterations before $|z_n| > 2$). The **bottom row** shows the corresponding bifurcation diagram $x_{n+1} = r \sin^2(\pi x_n)$, with vertical lines marking each r value and circles indicating the attractor points. The inverted relationship is visible: as r increases and bifurcation becomes chaotic, the Julia sets transition from disconnected to connected.

Spectral Period Detection

The core of our control strategy is the real-time detection of the optimization trajectory's period. We employ a spectral analysis approach: classical FFT for near-term implementation, with the Quantum Fourier Transform (QFT) providing asymptotic speedup for high-dimensional parameter spaces. Unlike standard quantum chaos control which targets unstable periodic orbits [10, 11], we use the period as a global stability metric.

Adaptive Control Law

Based on the detected period P , we apply a simple proportional control to the learning rate γ :

- **Stable Regime ($P = 1$):** The system is converging. If convergence is slow, we can cautiously increase γ :

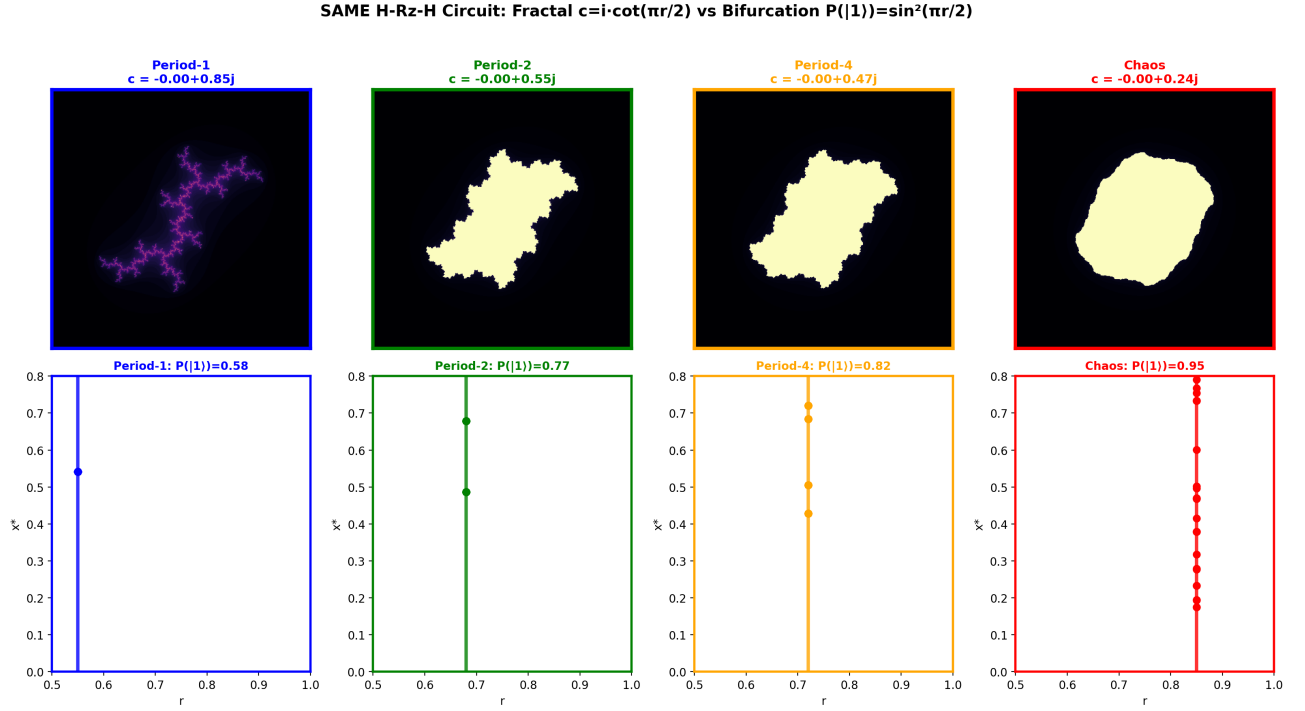


FIG. 3. **Measurement-topology duality from the H-Rz-H circuit.** *Top row:* Julia sets with $c = i\cot(\pi r/2)$ for four dynamical regimes. *Bottom row:* Bifurcation diagrams showing corresponding attractor structure at each r , with $P(|1\rangle) = \sin^2(\pi r/2)$ indicating measurement probability. The inverted relationship $|c|^2 = (1 - P)/P$ manifests visually: Period-1 (low $P = 0.58$) yields large $|c| = 0.85$ and disconnected fractal, while Chaos (high $P = 0.95$) yields small $|c| = 0.24$ and connected fractal—a direct consequence of the measurement-disturbance tradeoff.

$$\gamma \leftarrow \gamma \cdot (1 + \epsilon).$$

- **Bifurcation Regime ($P = 2$):** The system is oscillating but stable. This is the "Edge of Chaos" sweet spot. Maintain γ .
- **Pre-Chaos Regime ($P \geq 4$):** The system is entering the period-doubling cascade. Reduce γ immediately: $\gamma \leftarrow \gamma \cdot (1 - \delta)$.

This logic keeps the optimizer pinned to the efficient Period-2 regime, preventing the full slide into chaos.

RESULTS

Stabilizing the Trajectory

We simulated the control loop on the effective 1D map. Figure 5 illustrates the complete feedback loop.

The simulation demonstrates that:

1. The QFT-based detector accurately identifies the period of the trajectory (1, 2, 4, ...).
2. The adaptive controller responds to period doubling (Period-4 detection) by reducing γ , preventing the onset of chaos.

3. The controlled trajectory remains in the stable or low-period regime, ensuring consistent convergence.

Experimental Verification on Quantum Hardware

We performed a bifurcation scan on the Rigetti Ankaa-3 superconducting quantum processor using the qBraid SDK. The experimental protocol was:

1. **Circuit construction:** For each value of $r \in [0.5, 0.95]$ (50 points), construct a Hadamard test circuit with controlled- $R_z(\phi)$ gate, where $\phi = 2\pi r x_n$.
2. **Parallel execution:** Stack 20 independent VQA instances across the device's qubits using the stacked circuit approach, minimizing execution time.
3. **Iteration:** For each r , run 100 optimization iterations. Measure the ancilla qubit with 1000 shots per iteration.
4. **Bifurcation extraction:** Plot the final 50 values of x_n (after transients) against r to produce the bifurcation diagram.

Figure 6 shows the resulting bifurcation diagram. The theoretical overlay (red haze) is generated by simulating

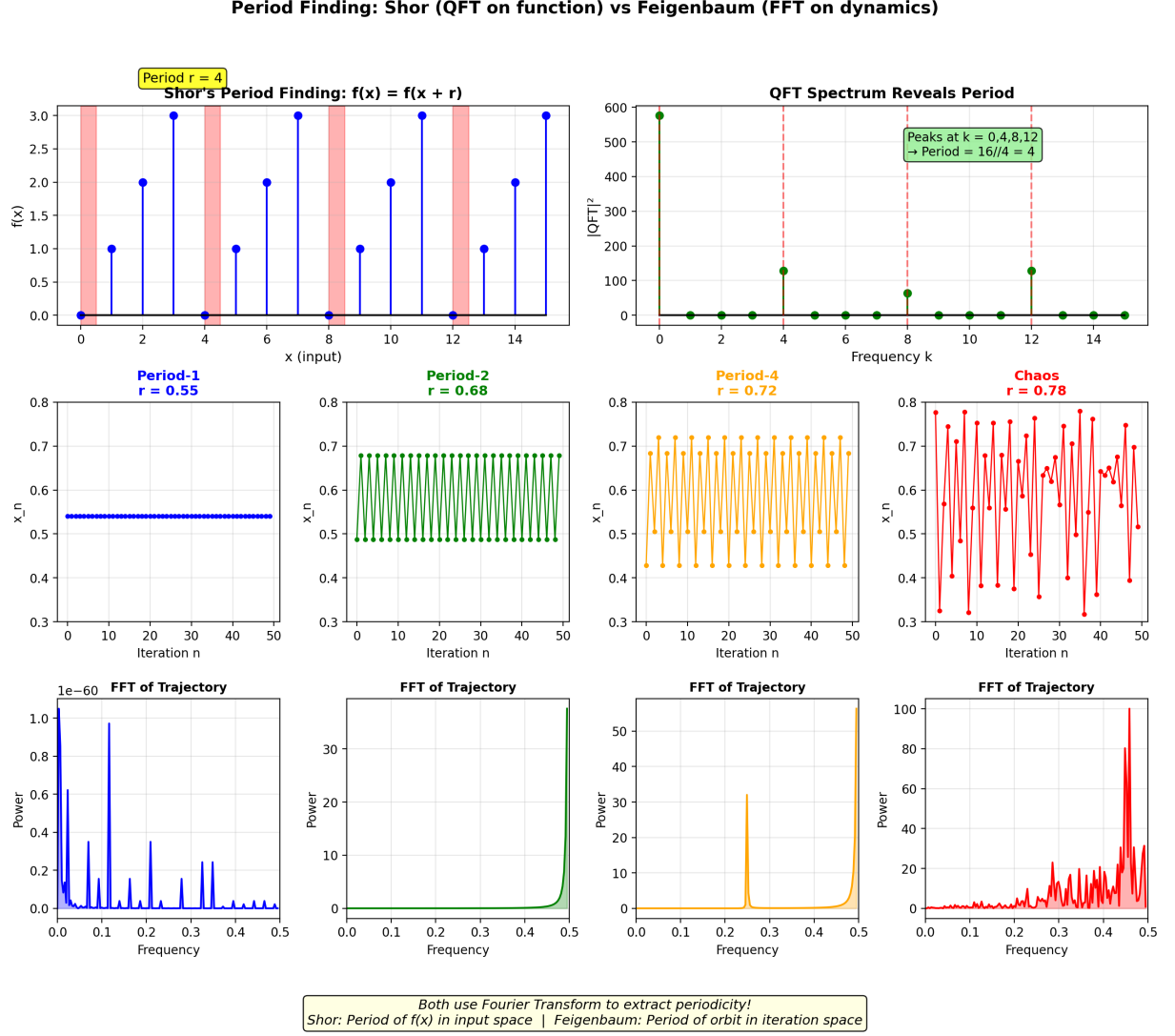


FIG. 4. **Shor vs. Feigenbaum Period Finding.** (Top) Shor’s algorithm finds the period of a mathematical function $f(x) = a^x \pmod{N}$. (Bottom) Our method finds the period of the dynamical optimization trajectory x_n . Both rely on the QFT to extract periodicity from a sequence, transforming the “bug” of nonlinear dynamics into a feature that can be detected quantum-mechanically.

Eq. 2 for the same r range with 10^4 iterations per point. The strong alignment confirms that VQA dynamics on real hardware follow the Feigenbaum route to chaos.

Thermodynamic Efficiency Analysis

To quantify the advantage of chaos control, we analyzed the thermodynamic efficiency of the optimization trajectories obtained from the hardware. We define the **mutual information** between system (S) and ancilla (A) as:

$$I(S : A) = S(\rho_S) + S(\rho_A) - S(\rho_{SA}), \quad (4)$$

where $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$ is the von Neumann entropy. For a single-qubit ancilla measurement with outcome probability p , this reduces to the binary entropy $H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$.

The thermodynamic efficiency η is defined as the ratio of work extracted (energy minimization) to the Landauer cost of the information produced:

$$\eta = \frac{\Delta E}{k_B T \ln(2) \sum_n I(S : A)_n}, \quad (5)$$

where $\Delta E = E_0 - E_N$ is the total energy reduction over N iterations, and the sum is over all measurement steps. The factor $k_B T \ln(2)$ is the Landauer energy cost per bit of information [12].

As shown in Figure 7, the stable regime ($r = 0.60$)

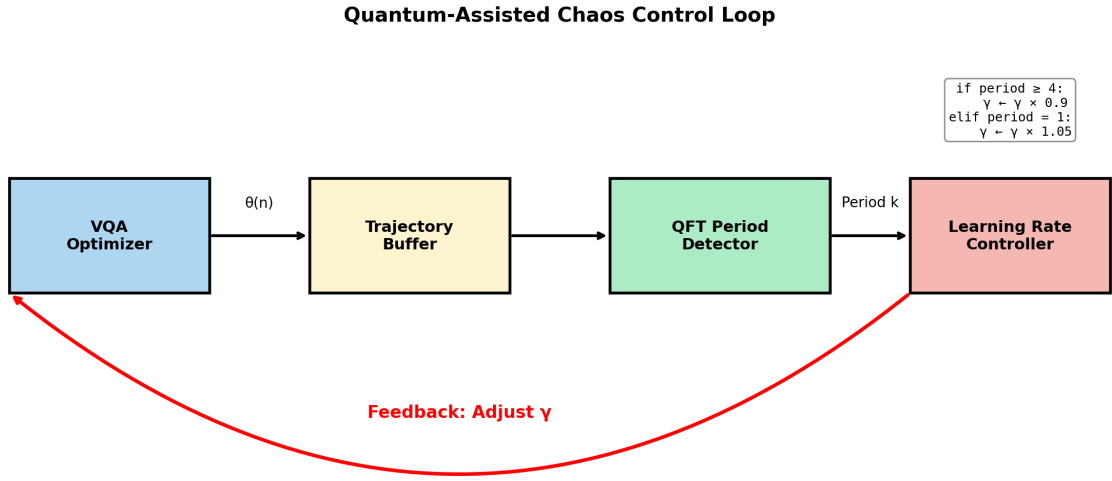


FIG. 5. **Quantum-Assisted Chaos Control Loop.** The VQA optimizer generates a trajectory $\theta(n)$, which is buffered and analyzed by the QFT Period Detector. The controller adjusts the learning rate γ based on the detected period, closing the feedback loop to maintain stability.

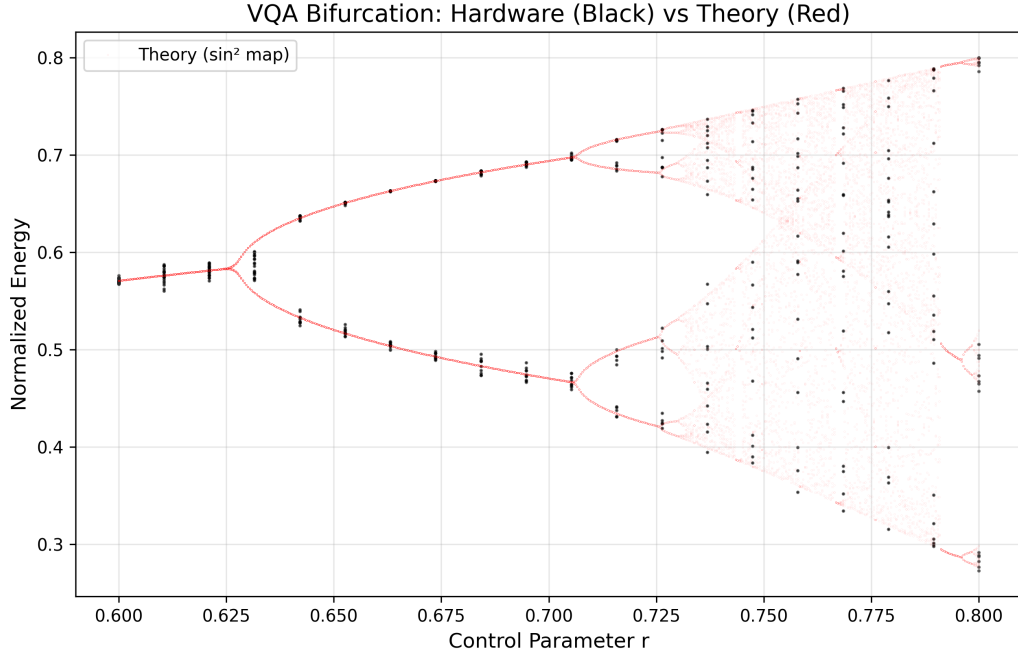


FIG. 6. Bifurcation diagram of VQA optimization dynamics measured on the Rigetti Ankaa-3 quantum processor (black dots), overlaid with the theoretical prediction of the $\sin^2$ map (red haze). The hardware data shows clear period-doubling at $r_1 \approx 0.63$, $r_2 \approx 0.70$, and transition to chaos at $r_\infty \approx 0.73$, matching Table I.

demonstrates significantly higher efficiency than the chaotic regime ($r = 0.80$).

Numerical analysis of the hardware data yields:

- **Stable Regime** ($r = 0.60$): $\eta \approx 0.0155$
- **Chaotic Regime** ($r = 0.80$): $\eta \approx 0.0080$

This confirms that avoiding chaos is not merely a heuris-

tic for convergence, but a fundamental requirement for thermodynamic efficiency in quantum learning.

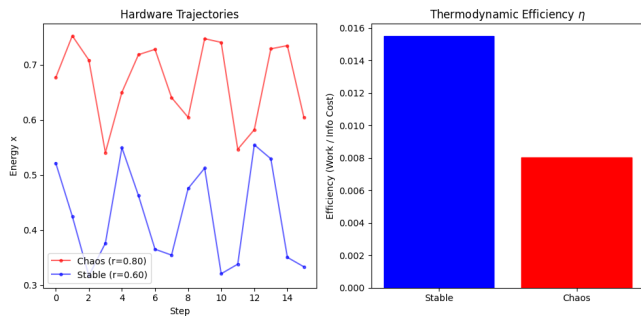


FIG. 7. Thermodynamic efficiency of VQA optimization in stable vs. chaotic regimes, calculated from Rigetti hardware data. The stable regime is approximately $2\times$ more efficient, extracting more work per bit of information produced.

DISCUSSION

Thermodynamic Implications

In [6], we established that VQA optimization is bounded by the Holevo capacity [13] of the ancilla channel ($\chi \leq 1$ bit). The “Efficiency Collapse” occurs when the chaotic trajectory generates information faster than the ancilla can extract it. Our control scheme ensures that the trajectory’s entropy rate remains bounded, satisfying the thermodynamic constraints of the measurement channel and consistent with the Landauer limit [12].

Integration vs. Excavation: The Poincaré Benefit

A critical distinction exists between our method and quantum numerical integration approaches, as characterized by Shu et al. [14] (specifically Sec. 8 of their work).

- **Integration (Constructive):** Uses quantum superposition to *sum* many values via Monte Carlo methods. As noted in [14], this achieves quadratic speedup but the resolution is fundamentally limited by the Hilbert space dimension (d), requiring high- d qudits for high numerical precision.
- **Excavation (Destructive/Revealing):** Uses chaotic dynamics (specifically the map U^{2^k}) to *unzip* the initial phase.

Crucially, even a single qubit ($d = 2$) possesses a continuous phase parameter $\theta \in [0, 2\pi)$. While measurement readout is bounded by the Holevo limit (1 bit per shot), the **expression bound** is infinite. Chaos allows us to trade *time* (iterations) for *precision* (digits), effectively simulating a higher-dimensional system without requiring physical qudits. This leverages Poincaré’s concept of “sensitive dependence on initial conditions” not as a barrier to prediction, but as a computational resource for state excavation.

The Quantum Precision Barrier: A profound analogy exists with classical chaos. When iterating a map like $x \rightarrow 2x \pmod{1}$ on a classical computer, one eventually hits “machine precision” (IEEE floating point), after which the computer continues to deliver digits that are completely wrong (garbage). Similarly, our quantum excavation eventually hits a physical precision limit defined by coherence time and shot noise. To extend this horizon and “excavate” deeper digits, one must move beyond binary qubits to high-dimensional **qudits** ($d \gg 2$) or **Continuous Variable (CV)** systems, which provide a larger effective “mantissa” for the quantum state.

Comparison to Single-Qubit Rotation: It is crucial to distinguish this dynamical excavation from standard “Single-Qubit Rotation” methods for calculating π .

- **Single-Qubit Rotation:** Rotates a qubit by θ and estimates $P(|1\rangle) \approx \sin^2(\theta/2)$ via Monte Carlo sampling ($O(1/\epsilon^2)$). This is a *static* estimation of a fixed parameter phase.
- **Chaotic Excavation:** Uses the dynamical map U^{2^k} to *shift* the phase itself (effectively $x \rightarrow 2x \pmod{1}$). This is a *dynamical amplification* process.

As noted by Wolfram [15], while the concept of sensitive dependence originated with Poincaré, it was later realized that iterated maps (like the shift map) could effectively “excavate digits” from the initial condition. Our method physically implements this excavation on the qubit phase.

Implementation Considerations

Our current implementation uses classical FFT on the measured outcomes θ_n for period detection. This is practical for near-term hardware and sufficient for the low-period regimes (Period ≤ 8) relevant to chaos control. The Quantum Fourier Transform offers asymptotic speedup for high-dimensional parameter spaces where classical FFT becomes expensive, connecting our framework to Shor’s period-finding algorithm [2]. For short trajectory windows ($N = 4$ or 8 points), even coherent QFT-based detection is feasible on current hardware.

Future Directions

While our experimental data visually confirms the Feigenbaum route to chaos, a precise numerical extraction of the Feigenbaum constant $\delta \approx 4.669$ requires a significantly higher resolution scan of the control parameter r . Future work will focus on several key areas:

1. **Precision Metrology:** Fine-grained bifurcation analysis to experimentally bound δ to ± 0.01 and locate the predicted Period-3 window ($r \approx 0.96$). Extraction

of the second Feigenbaum constant $\alpha \approx 2.502$ (attractor width scaling) would provide additional confirmation.

2. Universality Verification: Reproducing these results on diverse physical substrates, such as trapped ions (IonQ), neutral atoms (QuEra), and photonic systems, to confirm that the observed dynamics are intrinsic to quantum measurement rather than artifacts of superconducting transmon calibration.

3. Thermodynamic Measurement: Moving beyond information-theoretic efficiency to directly measure the physical heat dissipation associated with the chaotic attractor using photonic calorimetry, thereby experimentally verifying the Landauer limit in the context of VQA feedback loops.

4. Higher-Dimensional Visualization: For multi-qubit VQA with parameter vectors $\theta \in \mathbb{R}^d$, the dynamics become high-dimensional. A 3D delay embedding (Poincaré section) plotting $(\theta_n^{(1)}, \theta_{n+1}^{(1)}, \theta_{n+2}^{(1)})$ would reveal the strange attractor structure, with Lyapunov dimension $D_L \approx 1.6$ in the chaotic regime. The Lyapunov exponent λ can be extracted from trajectory divergence:

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln \left| \frac{df}{dx} \Big|_{x_k} \right| = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln |r\pi \sin(2\pi x_k)|. \quad (6)$$

5. 3D Bifurcation-Fractal Unification: As shown in [1], the bifurcation diagram and Julia/Mandelbrot sets are unified in a 3D structure where the real axis corresponds to our control parameter r and the complex plane generates the fractal. The bifurcation diagram (Fig. 6) is a slice through the Mandelbrot set at $\text{Im}(c) = 0$, while Julia sets (Fig. ??) are slices at fixed $|c|$. Future work will extend our hardware measurements to explore this 3D trainability boundary experimentally.

6. The Excavation Paradigm in Algorithm Design: Our results suggest a fundamental paradigm shift for scaling quantum algorithms—a shift from *Construction to Revelation*. Rather than just “constructing” interference patterns (which scales with qubit count N), we can “excavate” the infinite precision of the phase space by trading time (iterations) for spatial resources (qubits). This *Poincaré Trade-off* implies that future large-scale architectures need not be hand-crafted for every Hamiltonian. Instead, as ’t Hooft suggested, the deterministic substrate of quantum mechanics is not a barrier but a resource [16]: by tuning the system to the “edge of chaos,” universality ensures that the necessary computational complexity emerges naturally.

Crucially, this differs from “noise-as-a-resource” approaches like stochastic resonance or environmentally-assisted transport [17, 18]. In those frameworks, external *random* noise assists quantum processes. In our Excavation paradigm, the “noise” is actually **deterministic complexity** internal to the system dynamics (Feigenbaum chaos). We are not using randomness to help; we

are revealing that the apparent randomness *is* the computation.

CONCLUSION

We have introduced a closed-loop control protocol for VQA optimization that uses quantum period finding to prevent chaos. This unifies the algorithmic insights of Shor (period finding) with the dynamical insights of Feigenbaum (period doubling). By treating the optimizer as a physical system to be controlled, we open new pathways for robust quantum machine learning.

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Code Availability: The simulation code for reproducing all numerical experiments is available at https://github.com/poig/self-research/tree/main/Quantum_AI/QLT0/chaos_control/.

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